```
import pandas as pd
import seaborn as sns
```

```
tips = sns.load_dataset('tips')
tips.head()
```

	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4

Next steps: Generate code with `tips` New interactive sheet

```
titanic = pd.read_csv('train.csv')
```

```
flights = sns.load_dataset('flights')
flights.head()
```

	year	month	passengers
0	1949	Jan	112
1	1949	Feb	118
2	1949	Mar	132
3	1949	Apr	129
4	1949	May	121

Next steps: Generate code with `flights` New interactive sheet

```
iris = sns.load_dataset('iris')
```

```
iris.head()
```

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa

Next steps: Generate code with `iris` New interactive sheet

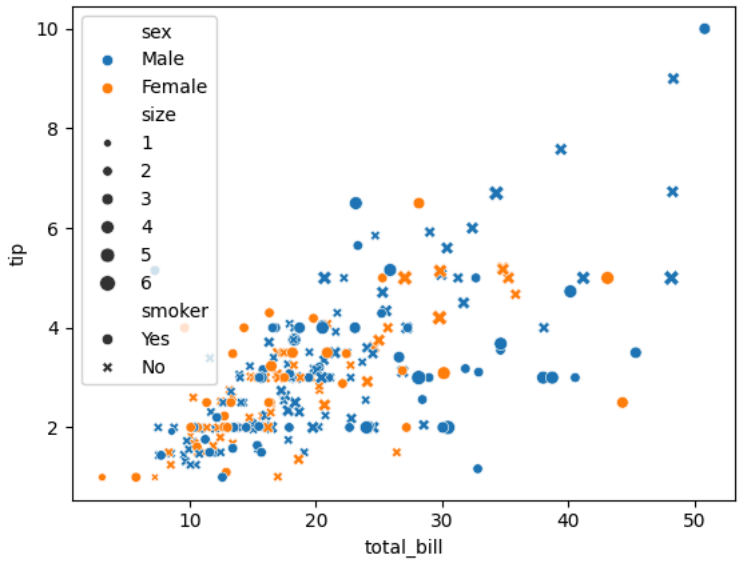
```
df.head()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S

Next steps: Generate code with `df` New interactive sheet

```
# scatterplot (numerical - numerical)
sns.scatterplot(x=tips['total_bill'], y=tips['tip'],hue=df['sex'], style=df['smoker'],size=df['size'])
```

```
<Axes: xlabel='total_bill', ylabel='tip'>
```



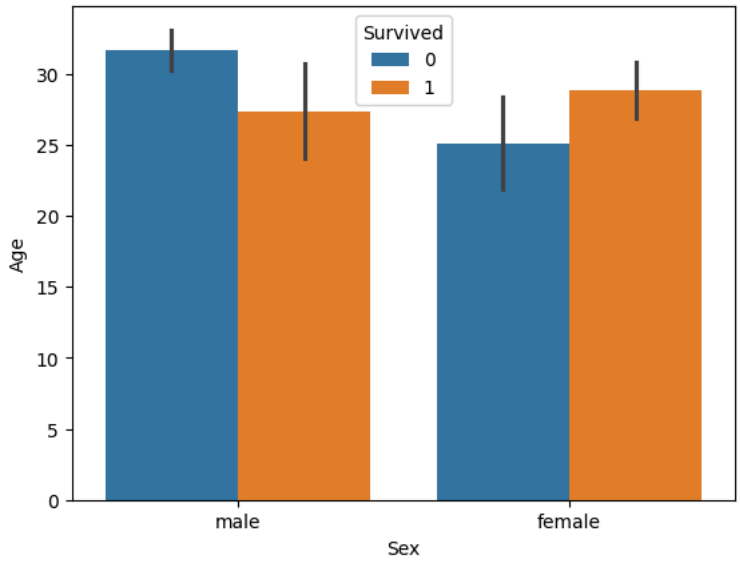

```
# Bar plot (numerical - categorical)
titanic.head()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S

Next steps: Generate code with `titanic` | New interactive sheet

```
sns.barplot(x=titanic['Sex'], y=titanic['Age'],hue=titanic['Survived'])
```

```
<Axes: xlabel='Sex', ylabel='Age'>
```




```
# 4 Displot(Numerical - categorical)
sns.distplot(titanic['Age'])
```

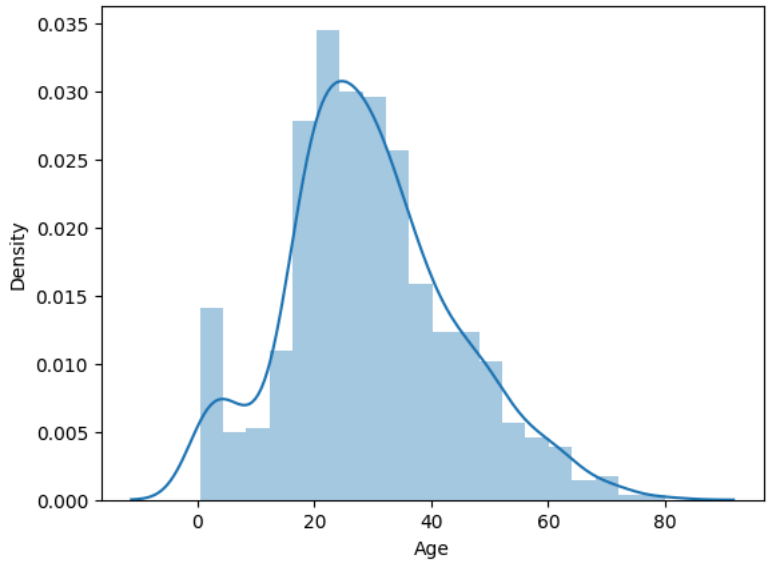
```
/tmp/ipykernel_939/1242488931.py:2: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.0.

Please adapt your code to use either `displot` (a figure-level function with
similar flexibility) or `histplot` (an axes-level function for histograms).

For a guide to updating your code to use the new functions, please see
https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751

  sns.distplot(titanic['Age'])
<Axes: xlabel='Age', ylabel='Density'>
```




```
sns.distplot(titanic[titanic['Survived']==0]['Age'],hist=False)
sns.distplot(titanic[titanic['Survived']==1]['Age'],hist=False)
```

```
/tmp/ipykernel_939/1261424998.py:1: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.0.

Please adapt your code to use either `displot` (a figure-level function with
similar flexibility) or `kdeplot` (an axes-level function for kernel density plots).

For a guide to updating your code to use the new functions, please see
https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751

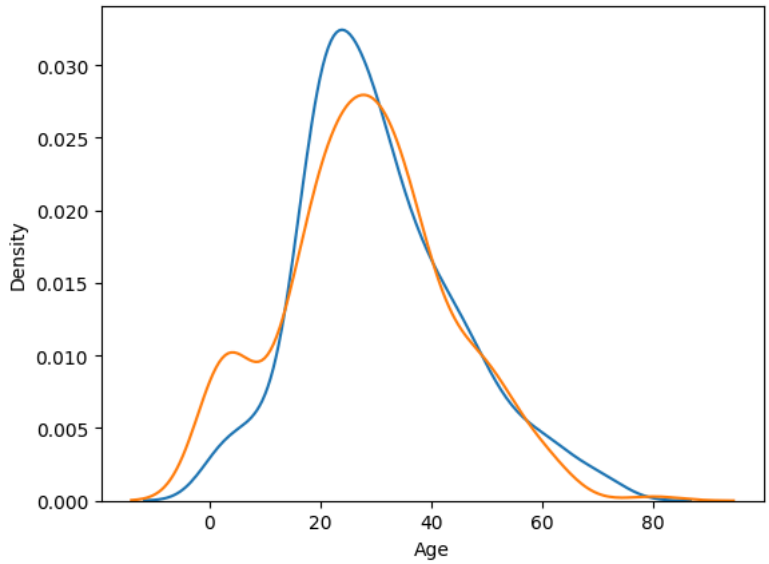
  sns.distplot(titanic[titanic['Survived']==0]['Age'],hist=False)
/tmp/ipykernel_939/1261424998.py:2: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.0.

Please adapt your code to use either `displot` (a figure-level function with
similar flexibility) or `kdeplot` (an axes-level function for kernel density plots).

For a guide to updating your code to use the new functions, please see
https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751

  sns.distplot(titanic[titanic['Survived']==1]['Age'],hist=False)
<Axes: xlabel='Age', ylabel='Density'>
```



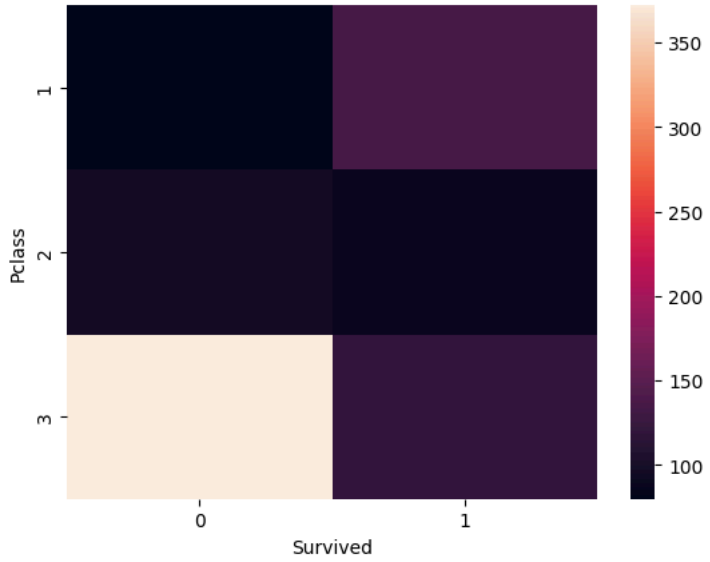

```
# Heatmap (Categorical - Categorical)
titanic.head(3)
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C

Next steps: Generate code with `titanic` New interactive sheet

```
sns.heatmap(pd.crosstab(titanic['Pclass'], titanic['Survived']))
```

```
<Axes: xlabel='Survived', ylabel='Pclass'>
```



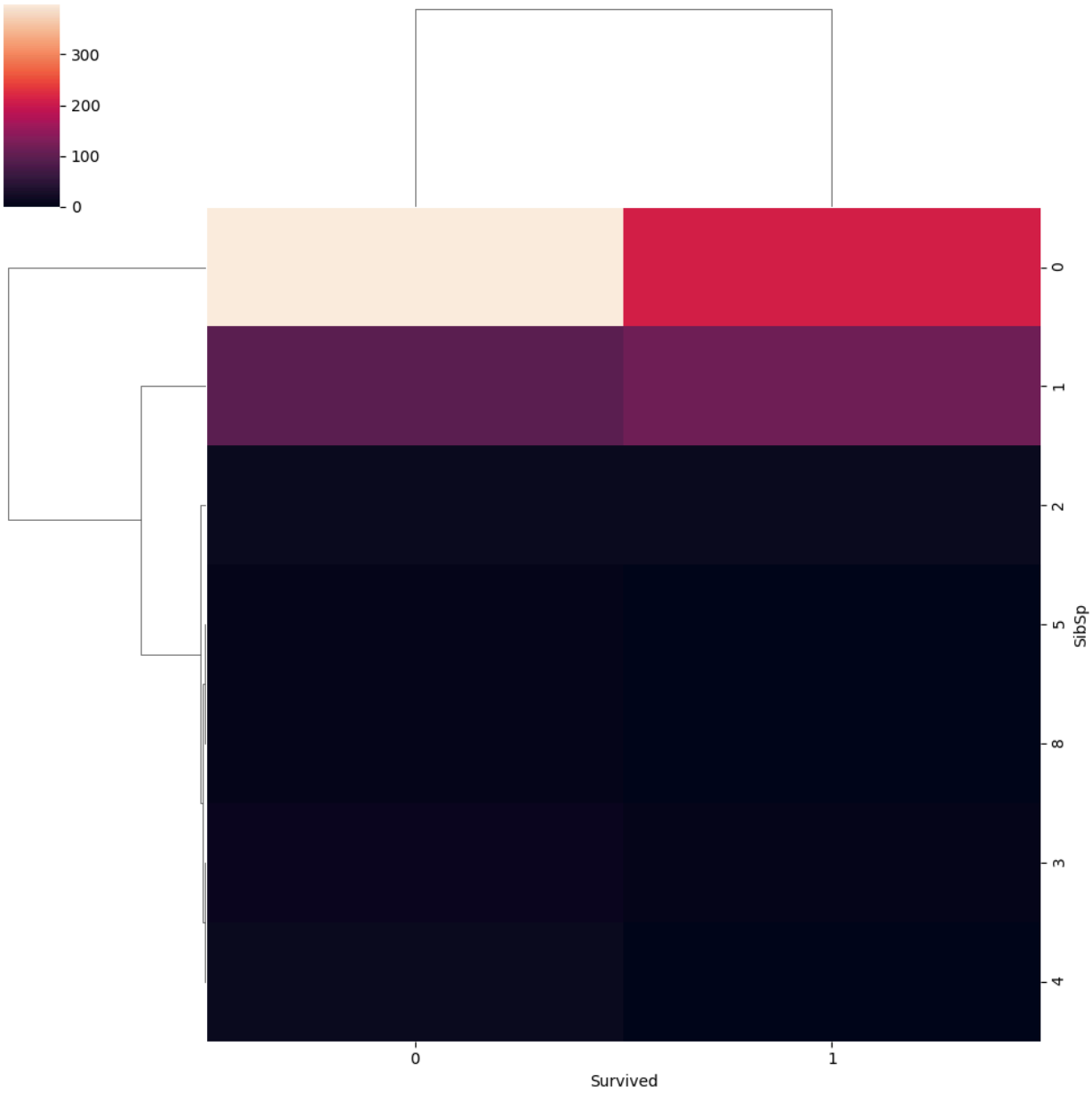

```
titanic.groupby('Survived')['Pclass'].mean()
```

Survived	Pclass
0	2.531876
1	1.950292

dtype: float64

```
# 6 cluster map (categorical - categorical)
sns.clustermap(pd.crosstab(titanic['SibSp'], titanic['Survived']))
```

```
<seaborn.matrix.ClusterGrid at 0x7a2f4d07b4d0>
```



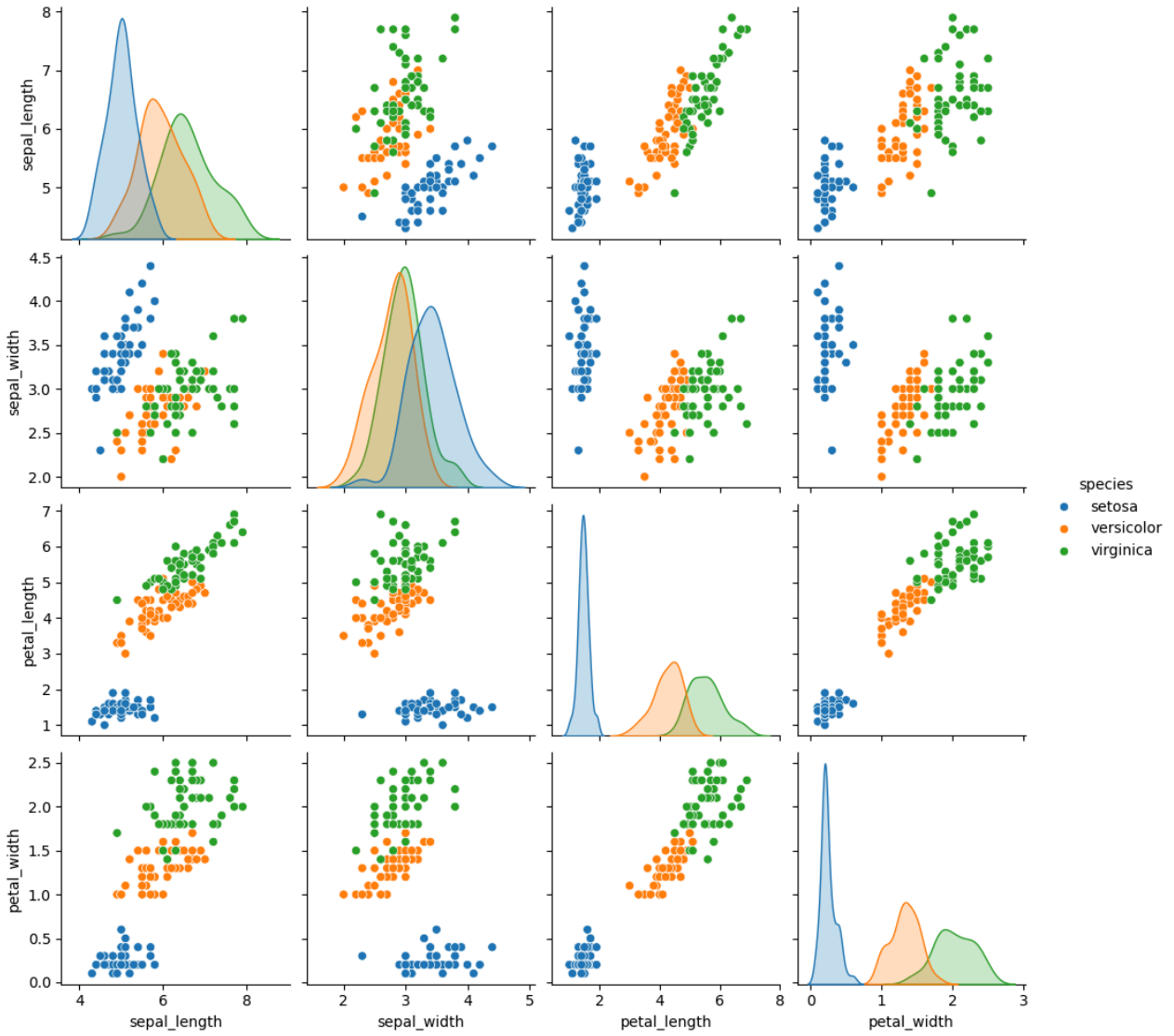

```
# pairplot
iris.head()
```

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa

Next steps: Generate code with `iris` New interactive sheet

```
sns.pairplot(iris,hue='species')
```

```
<seaborn.axisgrid.PairGrid at 0x7a2f4c752d50>
```




```
# lineplot ( Numercial- Numerical)
flights.head()
```

	year	month	passengers
0	1949	Jan	112
1	1949	Feb	118
2	1949	Mar	132
3	1949	Apr	129
4	1949	May	121

Next steps: Generate code with `flights` | New interactive sheet

```
new= flights.groupby('year')['passengers'].sum()
```

```
sns.lineplot(x=new.index, y=new.values)
```